Swipe

How Does The Internet Work ?

mdn web docs

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A simple network

When two computers need to communicate, you have to link them, either physically or wirelessly.

All modern computers can sustain any of those connections.

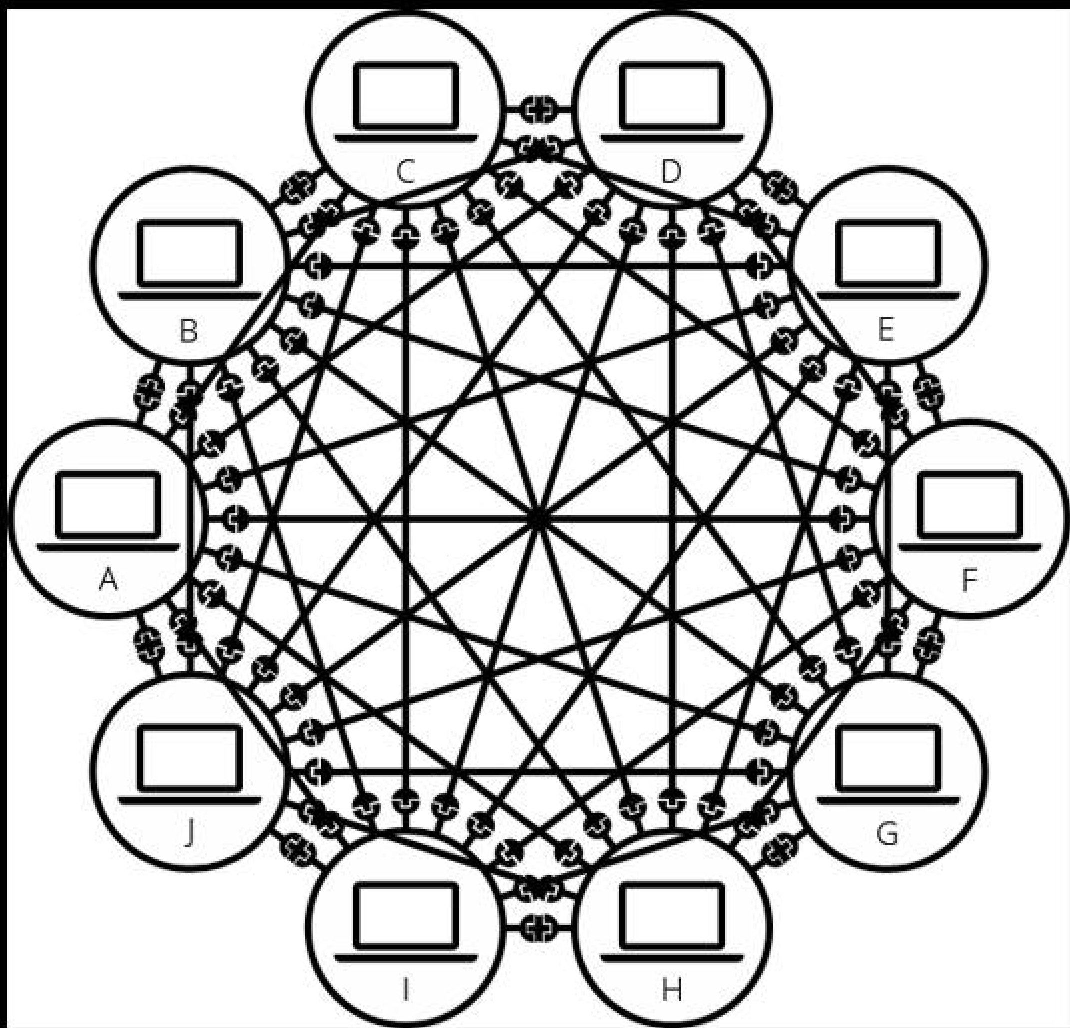
Example

Such a network is not limited to two computers. You can connect as many computers as you wish.

But it gets complicated quickly. If you're trying to connect, say, ten computers, you need 45 cables, with nine plugs per computer!

To solve this problem, each computer on a network is connected to a special tiny computer called a router.

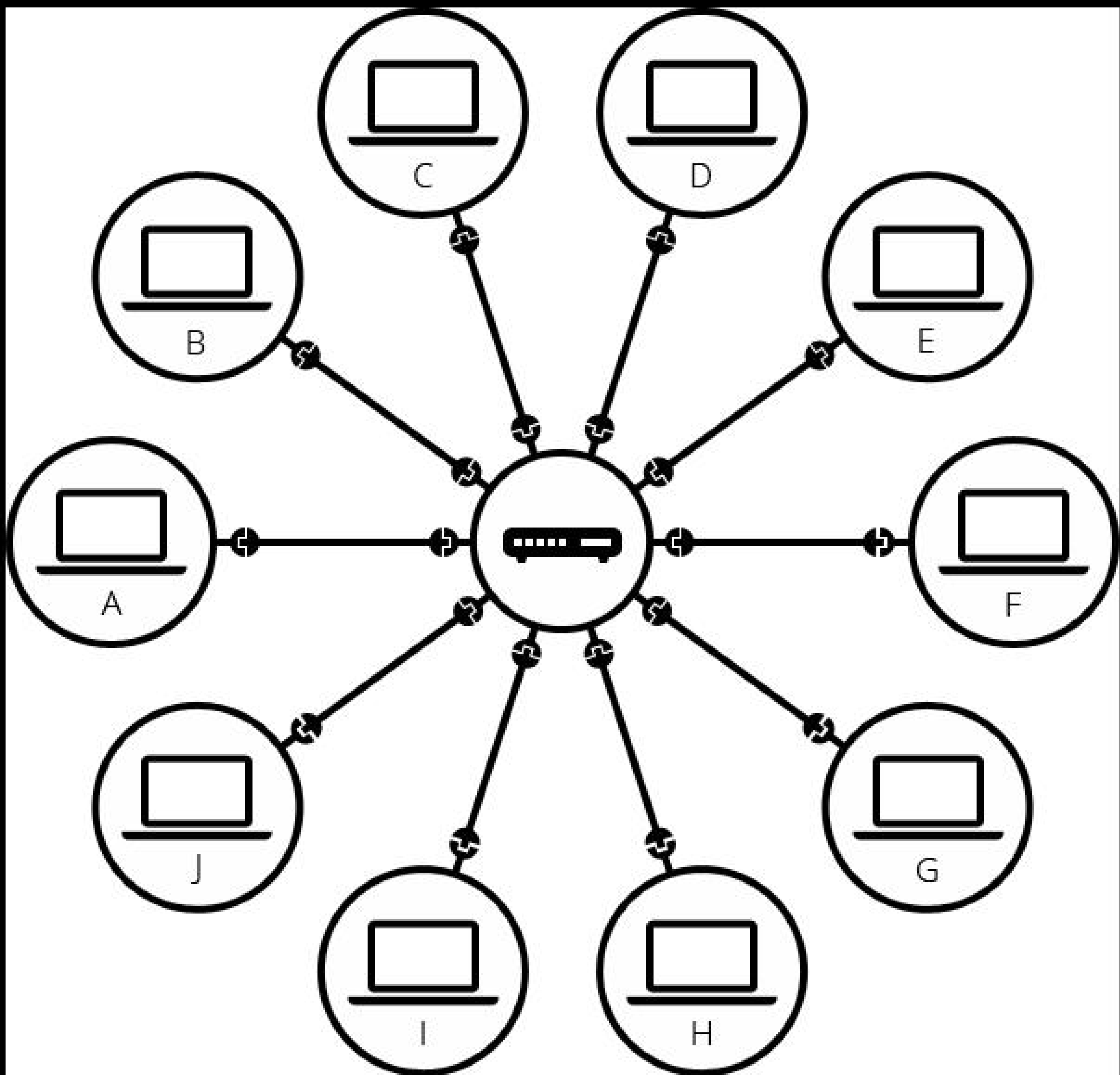
Example



This router has only one job: like a signaler at a railway station, it makes sure that a message sent from a given computer arrives at the right destination computer. To send a message to computer B, computer A must send the message to the router.

Once we add a router to the system, our network of 10 computers only requires 10 cables: a single plug for each computer and a router with 10 plugs.

Example



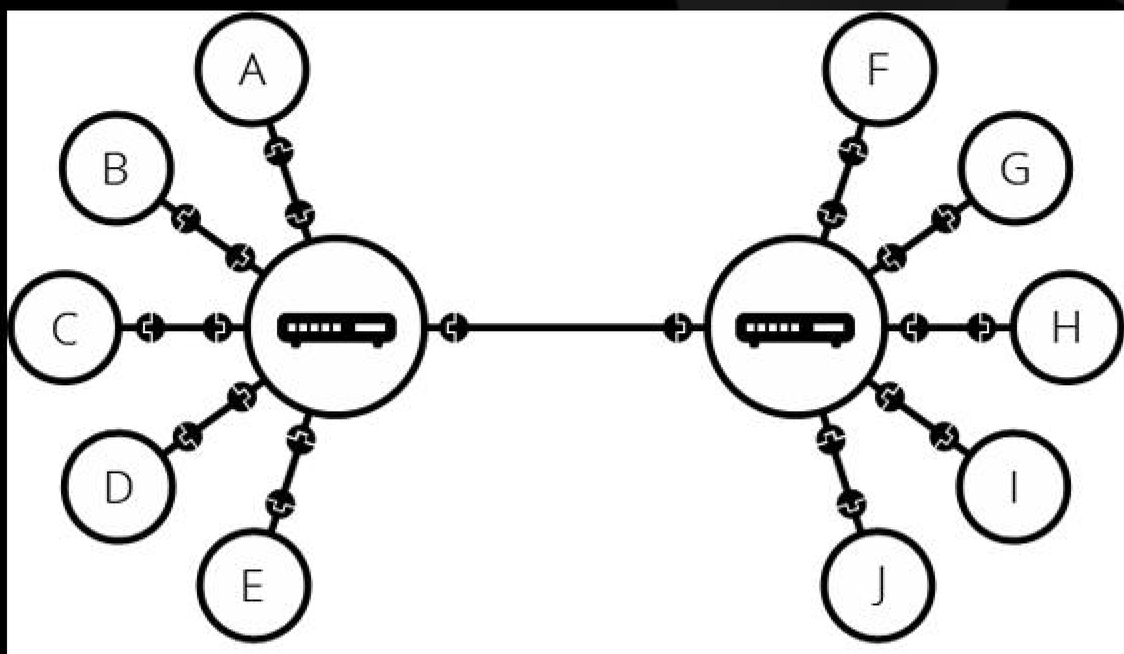
A network of networks

So far so good. But what about connecting hundreds, thousands, billions of computers?

Of course a single router can't scale that far,

but, as we know a router is a computer like any other, so we can connect two routers .

Example



By connecting computers to routers, then routers to routers, we are able to scale infinitely.

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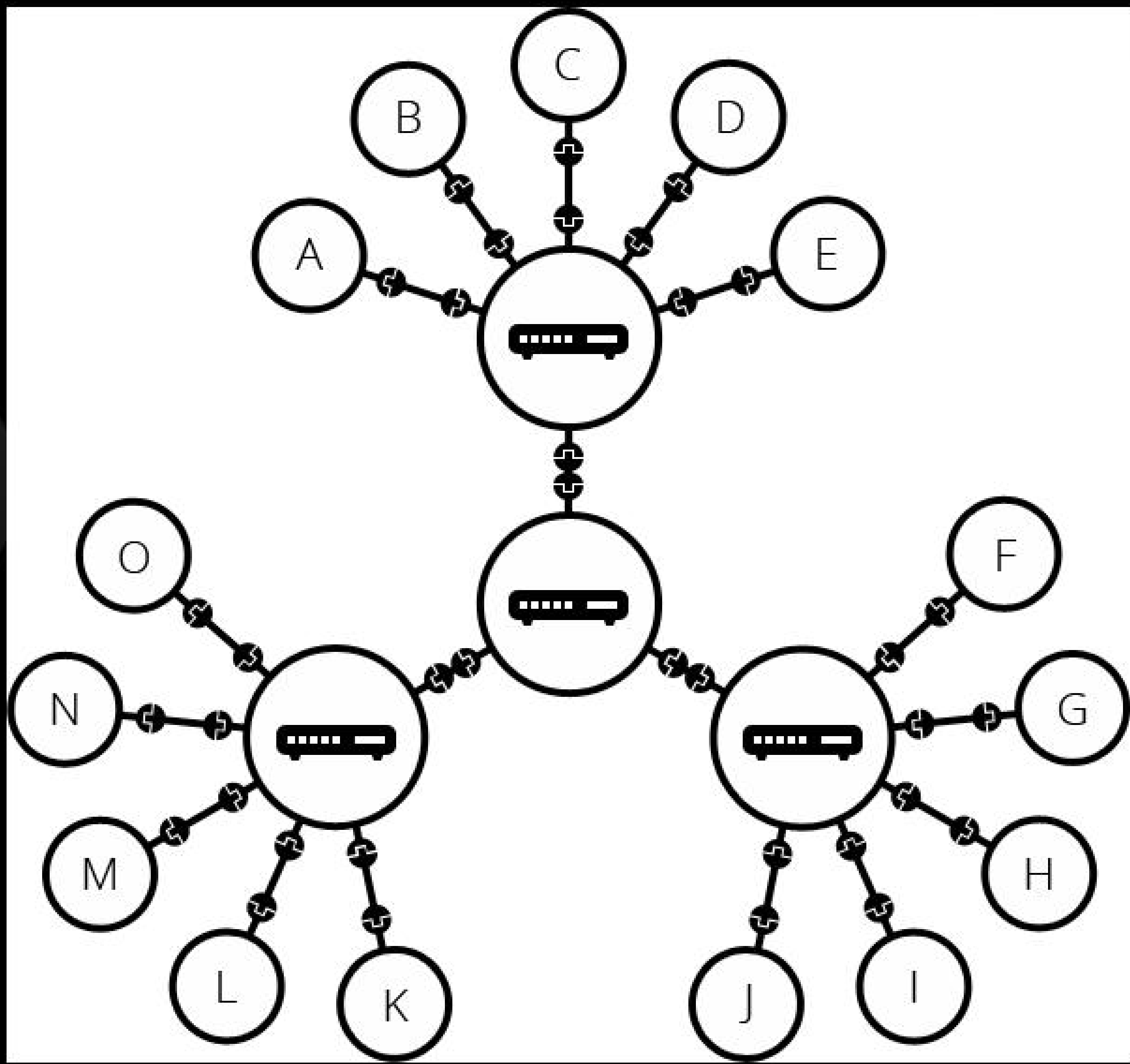
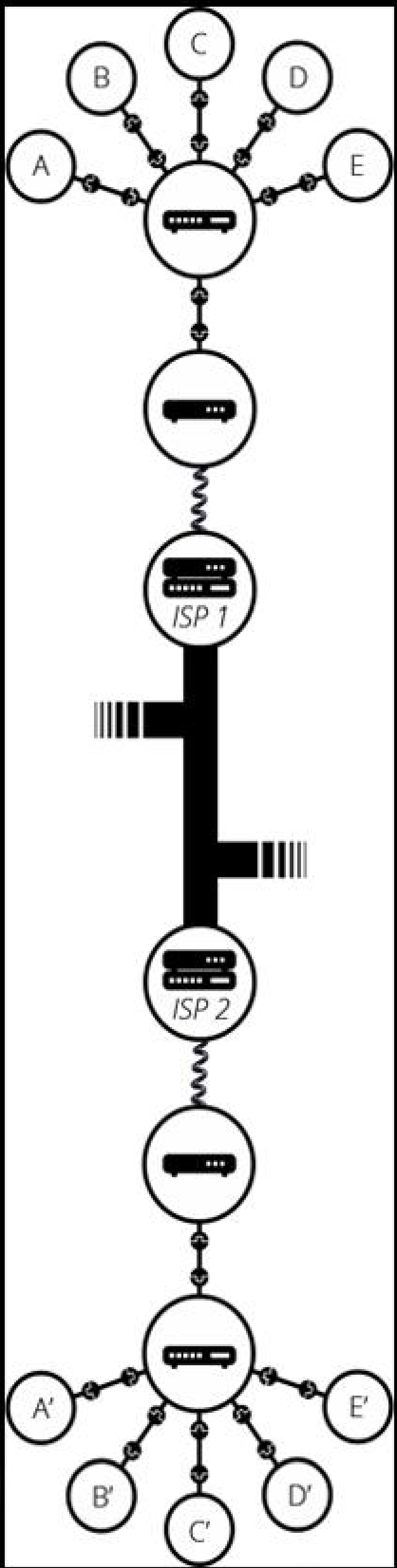
Example



Such a network comes very close to what we call the Internet,

but we're missing something.

The next step is to send the messages from our network to the network we want to reach. To do that, we will connect our network to an Internet Service Provider (ISP).



An ISP is a company that manages some special routers that are all linked together and can also access other ISPs' routers.

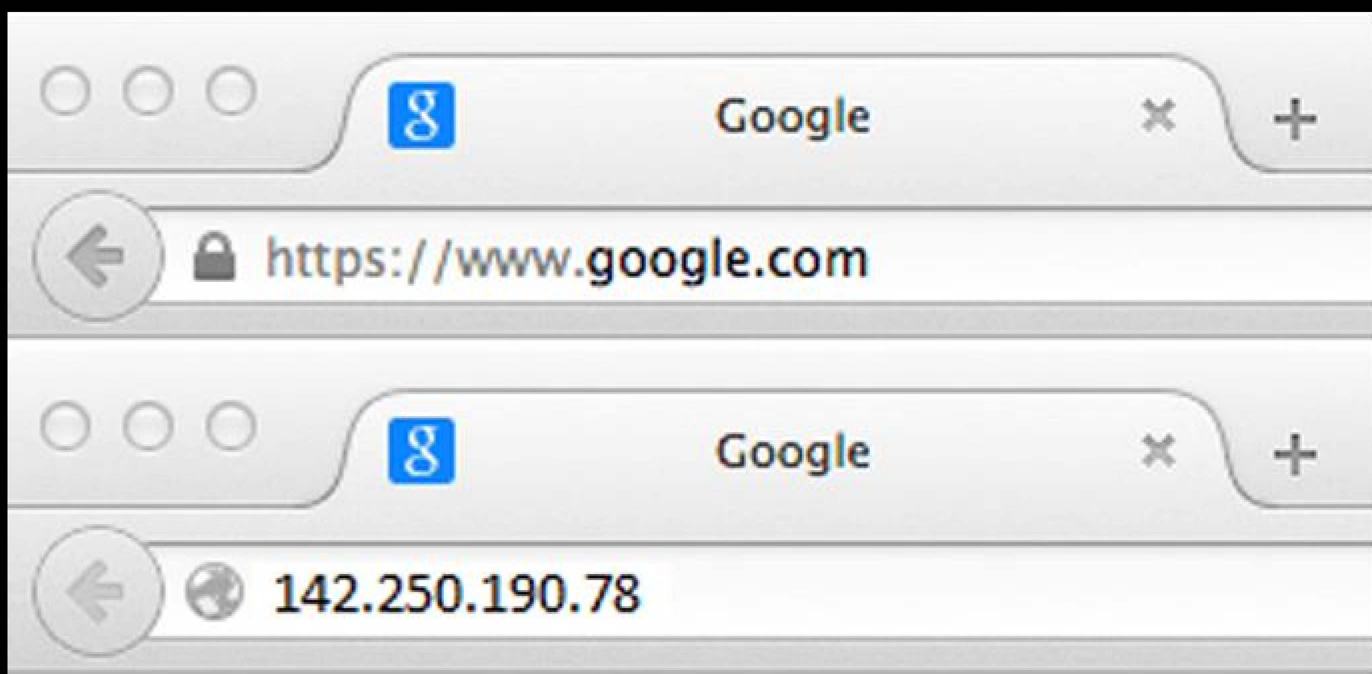
So the message from our network is carried through the network of ISP networks to the destination network. The Internet consists of this whole infrastructure of networks.

Example

Finding computers

If you want to send a message to a computer, you have to specify which one. Thus any computer linked to a network has a unique address that identifies it, called an "IP address" (Internet Protocol).

Example



It's an address made of a series of four numbers separated by dots, for example: 192.168.2.10.

That's fine for computers, but we human beings have a hard time remembering that sort of address.

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To make things easier, we can alias an IP address with a human-readable name called a domain name.

For example (at the time of writing; IP addresses can change) google.com is the domain name used on top of the IP address 142.250.190.78.

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